# Современные архитектуры нейронных сетей

## Диффузионные сети (продолжение)

# DDIM

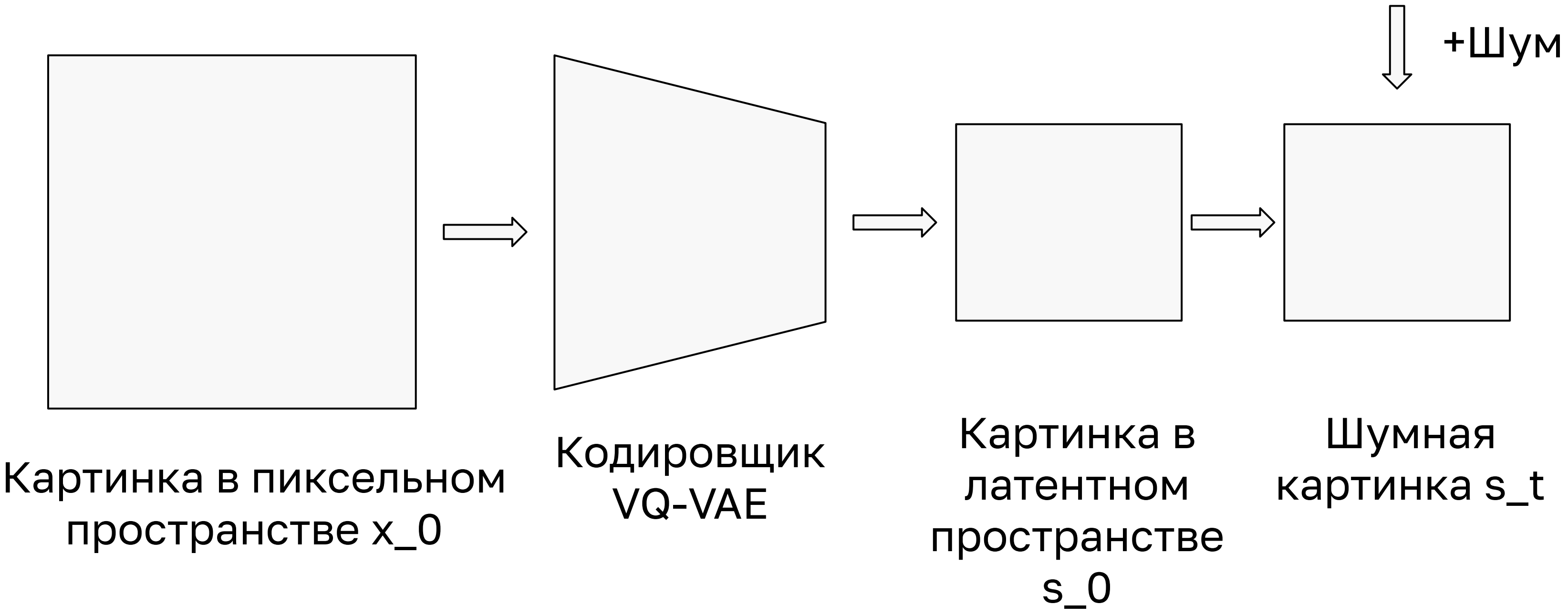
ИТМО

$$\boldsymbol{x}_{t-1} = \sqrt{\alpha_{t-1}} \underbrace{\left( \frac{\boldsymbol{x}_t - \sqrt{1-\alpha_t}\epsilon_\theta^{(t)}(\boldsymbol{x}_t)}{\sqrt{\alpha_t}} \right)}_{\text{`` predicted } \boldsymbol{x}_0 \text{''}} + \underbrace{\sqrt{1-\alpha_{t-1}-\sigma_t^2} \cdot \epsilon_\theta^{(t)}(\boldsymbol{x}_t)}_{\text{``direction pointing to } \boldsymbol{x}_t \text{''}} + \underbrace{\sigma_t \epsilon_t}_{\text{random noise}}$$

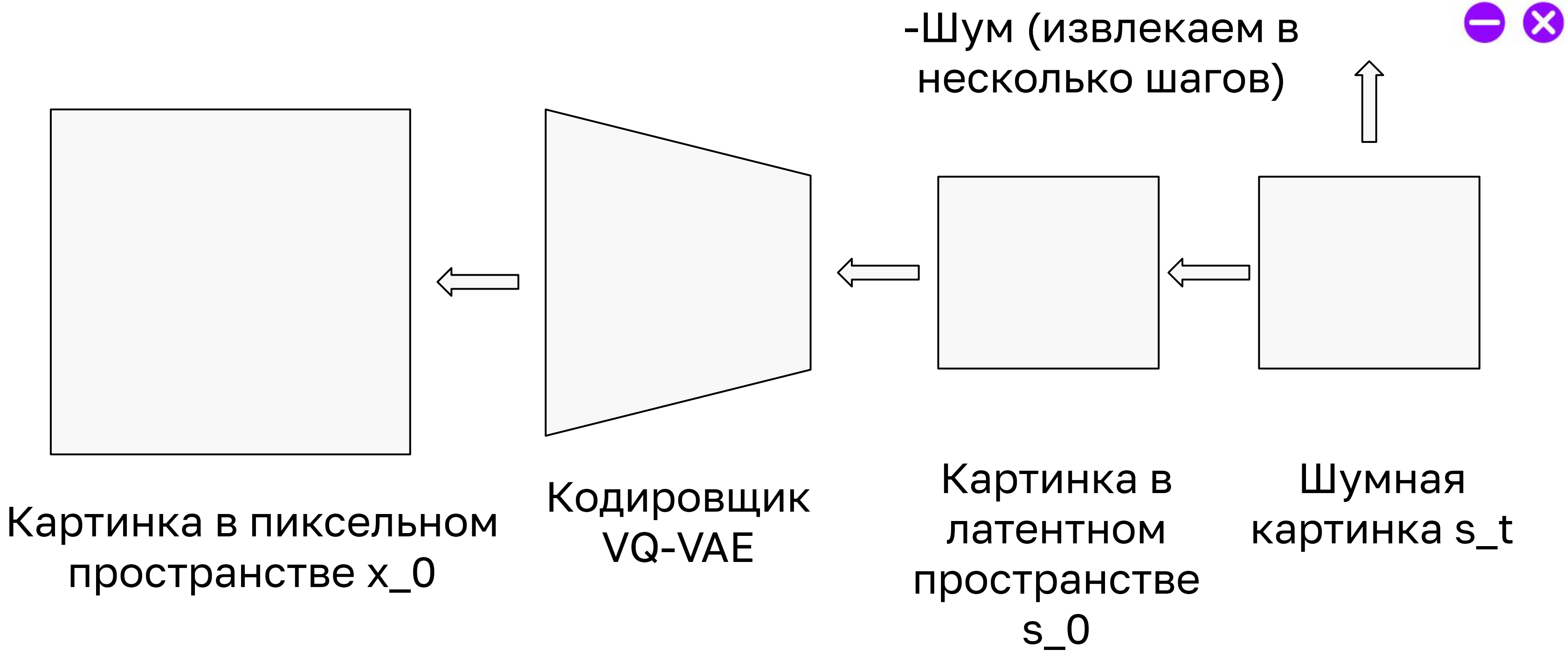
# Латентная диффузия

Какая проблема есть у DDPM?

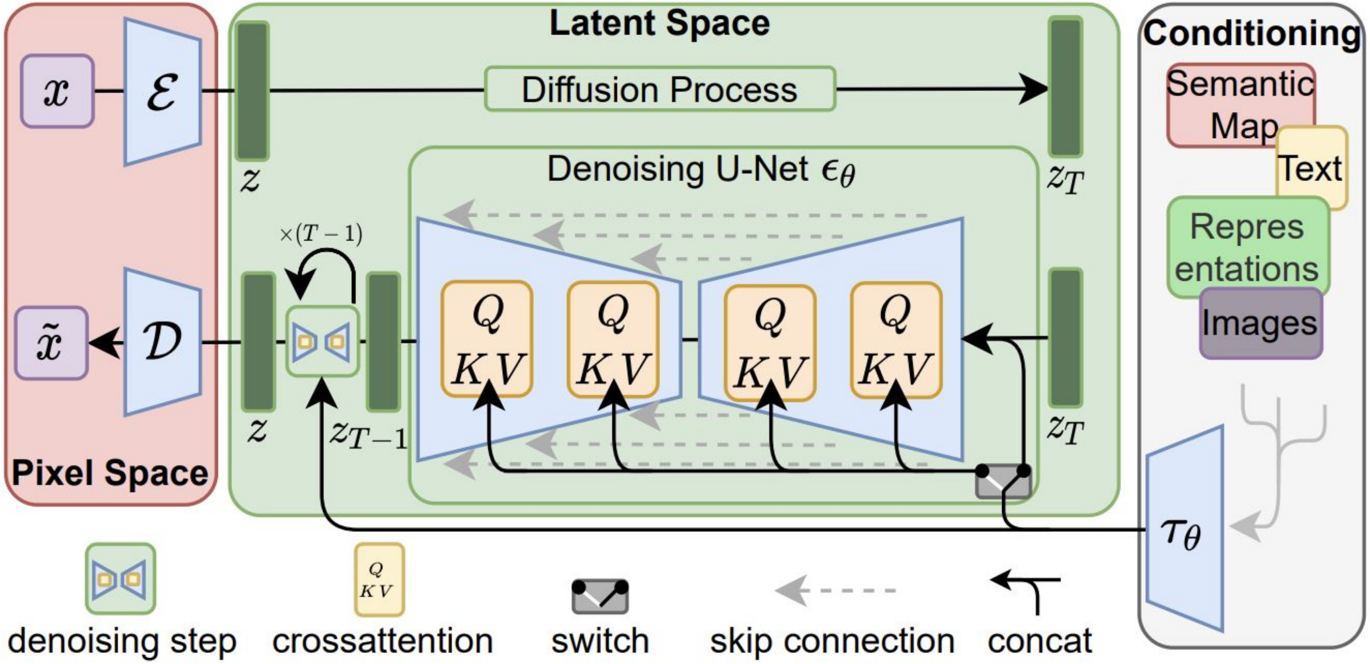
# Латентная диффузия (прямая диффузия)



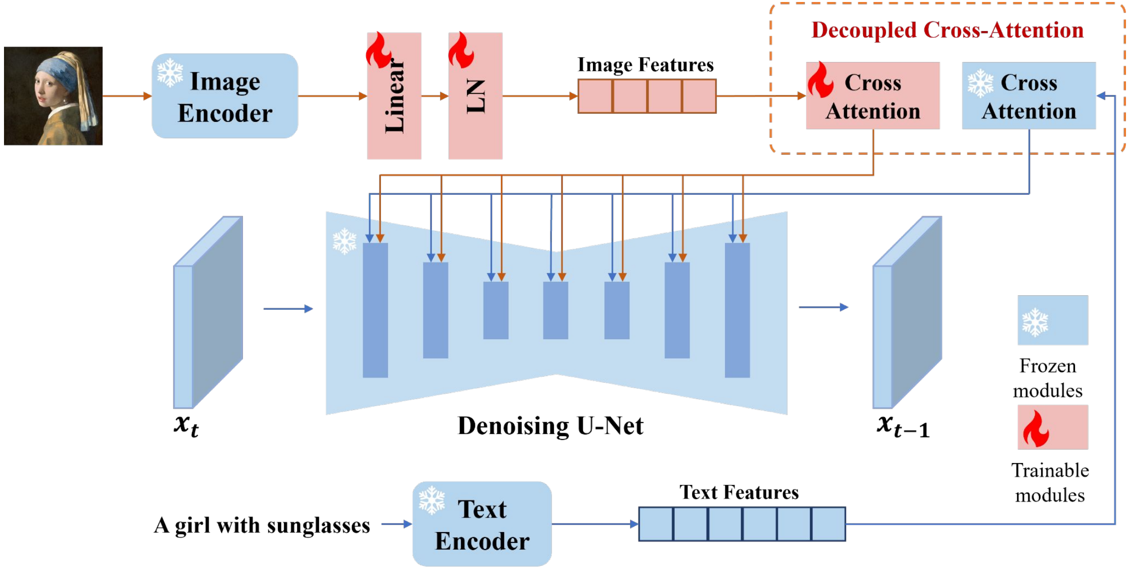
# Латентная диффузия (обратная диффузия)



# Латентная диффузия (Stable Diffusion)



# Увеличиваем контроль над генерацией (Адаптеры)



# Увеличиваем контроль над генерацией (Адаптеры)

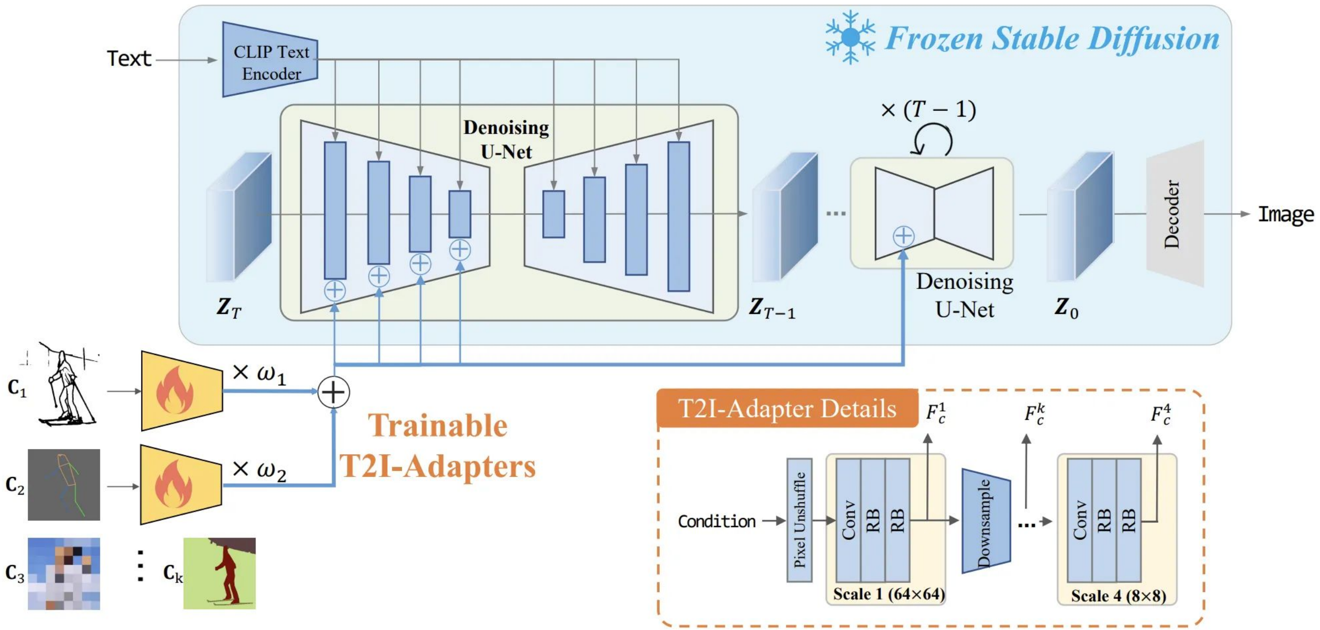
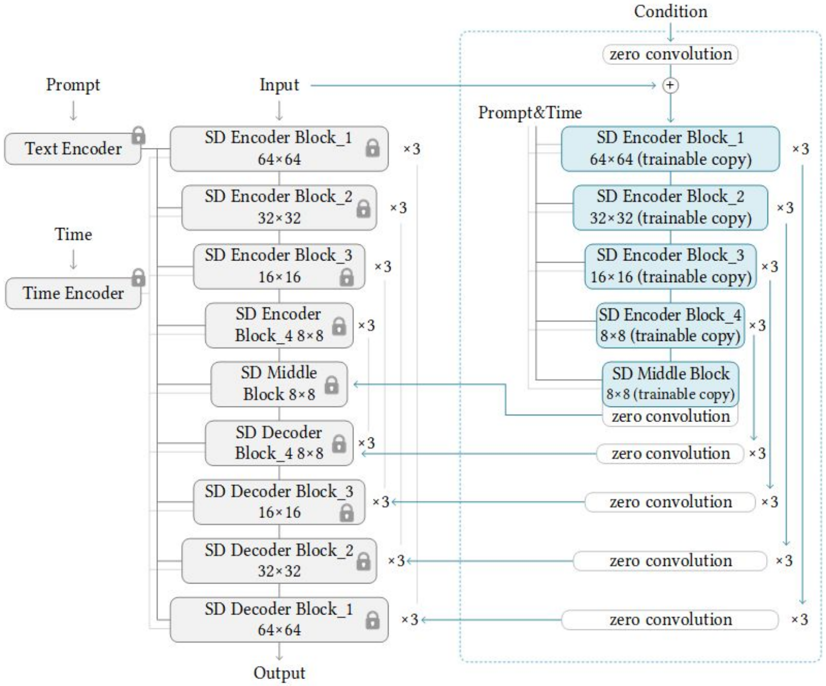


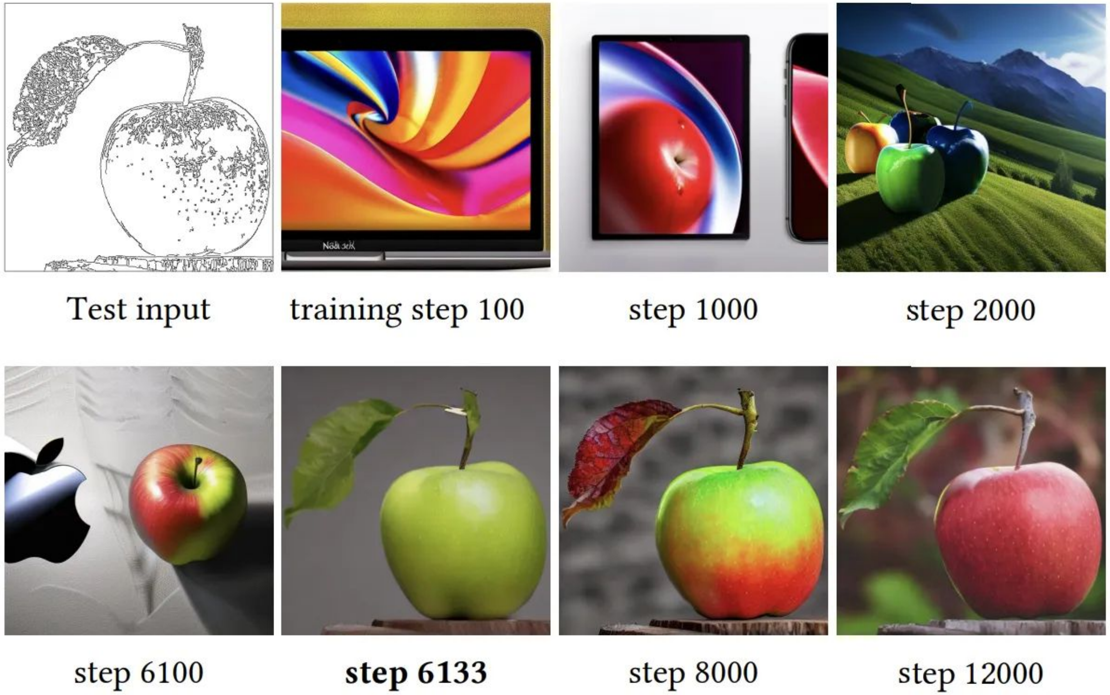

Figure 3. The overall architecture is composed of two parts: 1) a pre-trained stable diffusion model with fixed parameters; 2) several T2I-Adapters trained to align internal knowledge in T2I models and external control signals. Different adapters can be composed by directly adding with adjustable weight $\omega$. The detailed architecture of T2I-Adapter is shown in the lower right corner.

# Увеличиваем контроль над генерацией (ControlNet)

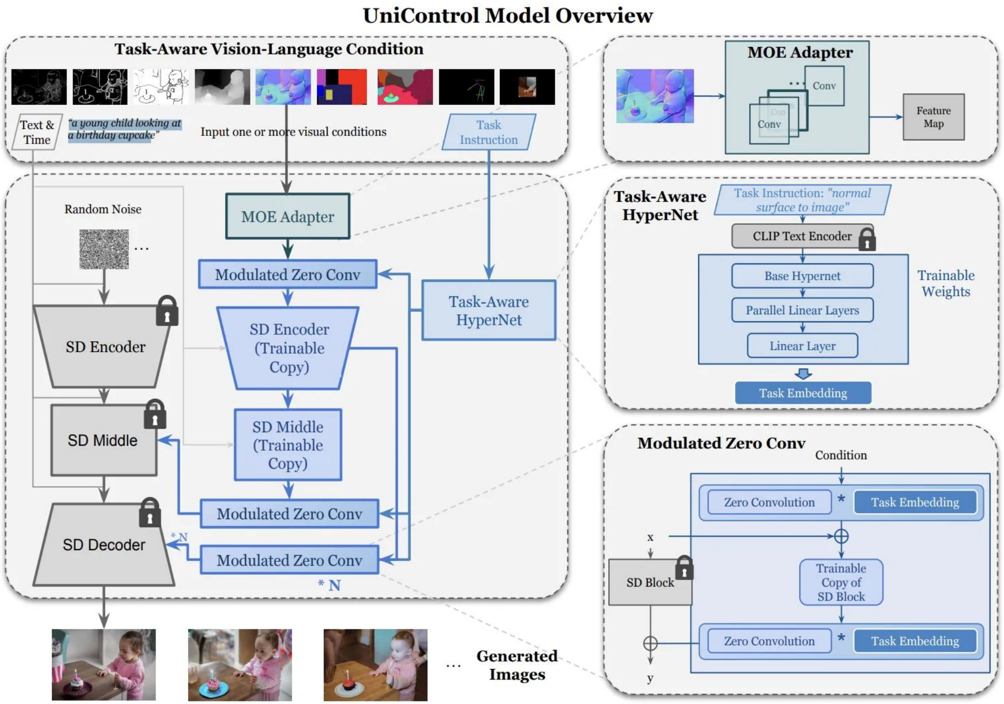



(a) Stable Diffusion (b) ControlNet

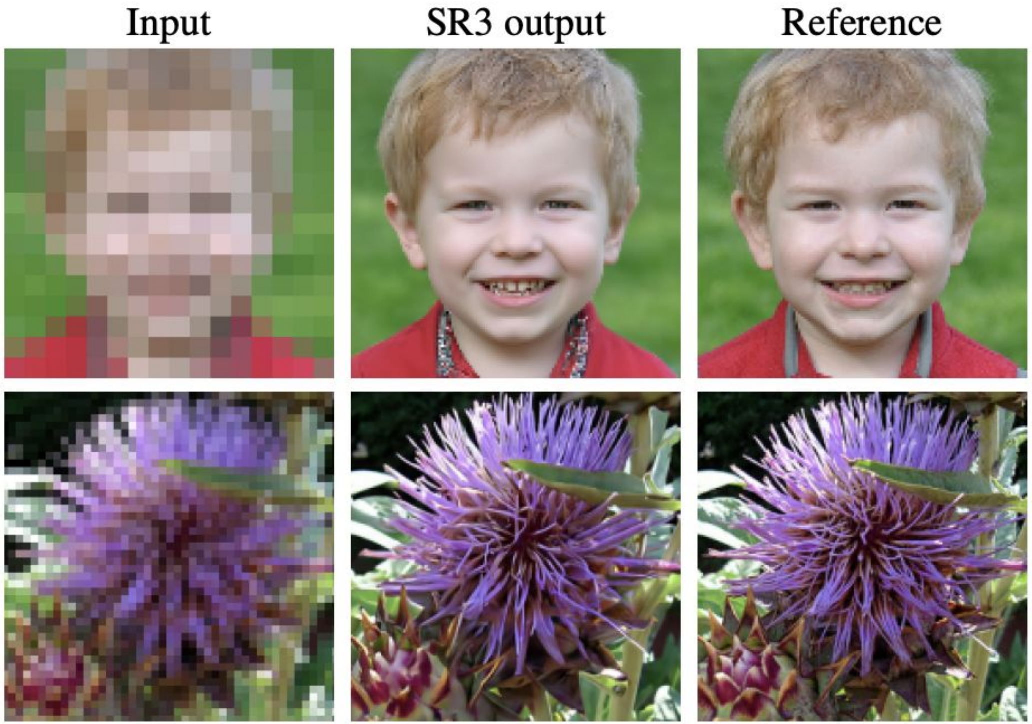
# Увеличиваем контроль над генерацией (ControlNet)



# Увеличиваем контроль над генерацией (UniControl)



# Diffusion Super Resolution



# Diffusion Transformer (DiT)

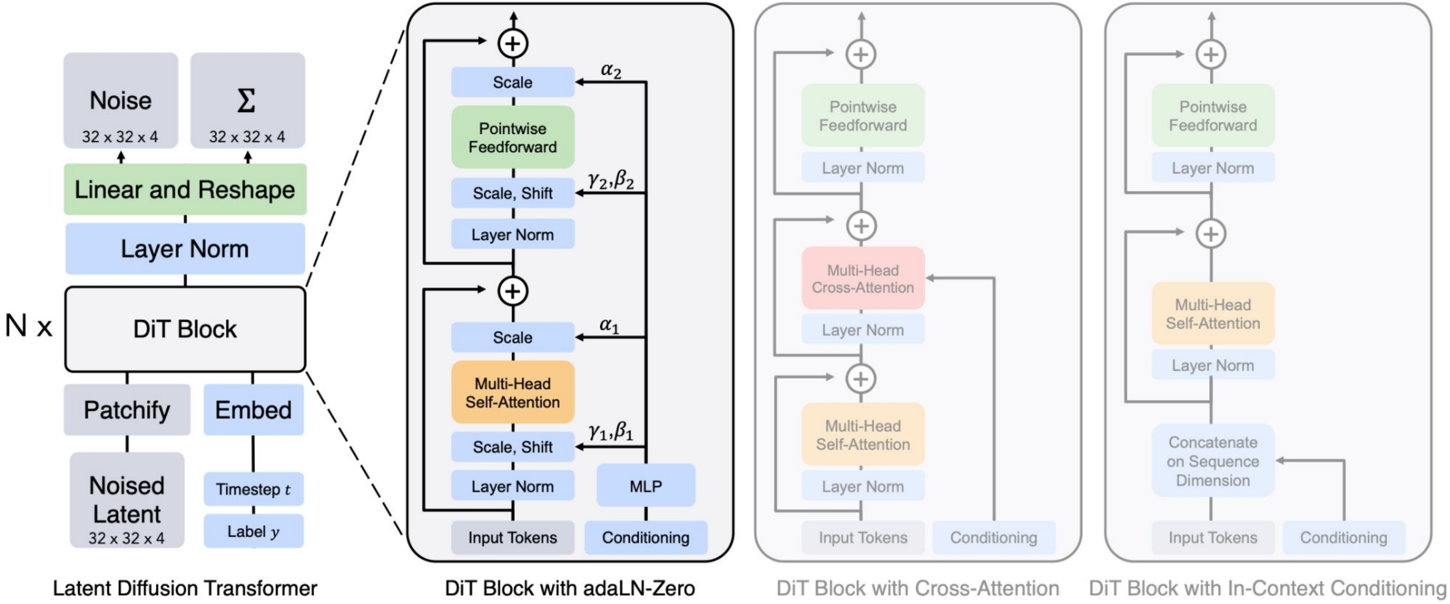



Figure 3. **The Diffusion Transformer (DiT) architecture.** *Left:* We train conditional latent DiT models. The input latent is decomposed into patches and processed by several DiT blocks. *Right:* Details of our DiT blocks. We experiment with variants of standard transformer blocks that incorporate conditioning via adaptive layer norm, cross-attention and extra input tokens. Adaptive layer norm works best.

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